The differences between disaster recovery and backup: Why you need both

Backup	Disaster recovery
Backup is the process of making an extra copy (or multiple copies) of data. You back up data to protect it.	Disaster recovery, on the other hand, refers to the plans and processes for quickly re-establishing access to applications, data and IT resources after an outage.
You might need to restore backup data if you encounter an accidental deletion, database corruption, or problem with a software upgrade.	The disaster recovery plan might involve switching over to a redundant set of servers and storage systems until your primary data centre is functional again.

before 11:00 a.m. Data loss will span only one hour, between 11:00 a.m. and 12:00 p.m. (noon).

- **Failover** is the disaster recovery process of automatically offloading tasks to backup systems in a way that is seamless to users. You might fail over from your primary data centre to a secondary site, with redundant systems that are ready to take over immediately.
- **Failback** is the disaster recovery process of switching back to the original systems. Once the disaster has passed and your primary data centre is up and running, you should be able to fail back seamlessly as well.
- **Restore** is the process of transferring backup data to the primary system or data centre. The restore process is generally considered part of backup rather than disaster recovery.

A company typically decides on an acceptable RTO and RPO based on the financial impact to the business when systems are unavailable. The company determines the financial impact by considering many factors, such as the loss of business and damage to its reputation due to downtime and the lack of systems availability.

IT organisations then plan solutions to provide cost-effective system recovery based on the RPO and on the service level established by the RTO.

The key capabilities of an efficient IT disaster recovery plan should be:

- The life cycle of the recovery, covering the total provision, monitoring, validating, testing, recovery and reporting processes
- Close integration with the network, OS and replication technologies to enable the automation of the monitoring, discovery and recovery processes
- Close integration with a virtual infrastructure to ensure automation of virtual machines for recovery
- Application awareness to ensure best practice templates for various enterprise applications

A disaster recovery plan (DRP) and its importance

To ensure that your systems, data and personnel are protected, and that your business can continue to operate in the event of an actual emergency or disaster, companies can follow the guidelines given below to create a disaster plan.

- **Inventory hardware and software:** Your DR plan should include a complete inventory of [hardware and] applications in priority order.
- **Define your tolerance for downtime and data loss:** Evaluate what an acceptable recovery point objective (RPO) and recovery time objective (RTO) is for each set of applications.
- **Lay out who is responsible for what – and identify backup personnel:** All disaster recovery plans should clearly define the key roles, responsibilities and parties involved during a DR event.
- **Create a communication plan:** Communication is critical when responding to and recovering from any emergency, crisis or disaster.

Top open source FOSS backup and DR tools

AMANDA

AMANDA stands for Advanced Maryland Automatic Network Disk Archiver. It is a simple backup solution that allows the IT administrator to set up a single master backup server to back up multiple hosts over the network to media like tapes, hard disk drives, etc.

AMANDA uses native utilities and formats, and can back up a large number of servers and workstations running multiple versions of Linux or UNIX. It uses a native Windows client to back up Microsoft Windows desktops and servers.

Rsync

A case study of Rsync is given at *https://www.osradar.com/use-rsync-to-set-up-remote-disaster-recovery-backup-system/*. To ensure data security, an enterprise felt the need for a remote disaster system. So, every morning at 3.00 a.m., data from the site is backed up to a remote server using Rsync. Because the volume of data is very large, the daily backup can only be incremental. In the case of a failure on site, the backed up data can be restored, helping the enterprise to a great extent.

Box Backup

This is a totally automatic, open source online backup system, using which the